

KALPANA BANNAD

+91 7829151853 | gouribannad@gmail.com | [LinkedIn](#) | [GitHub](#) | Vijayapur, Karnataka, India

PROFESSIONAL SUMMARY

Electronics and Communication Engineering student specializing in VLSI Design and AI/ML, with hands-on experience in RTL design, verification, and end-to-end system development. Proficient in hardware description languages such as Verilog and VHDL, with working knowledge of industry tools including Cadence Virtuoso. Strong foundation in digital design, timing analysis, and verification concepts, along with experience in modern software and AI frameworks such as TensorFlow, OpenCV, Flask, and REST APIs. Highly motivated to contribute to front-end VLSI and interdisciplinary engineering roles.

EDUCATION

BLDEA's V. P. Dr. P. G. Halakatti College of Engineering and Technology, Vijayapur

CGPA: 9.2 / 10

Bachelor of Engineering in Electronics and Communication Engineering

Expected Graduation: 2027

TECHNICAL SKILLS

Languages: Python, C, Verilog,

VLSI / Hardware: RTL Design, Cadence Virtuoso, FPGA, Embedded Systems, Testbench Verification (Icarus Verilog)

AI / Machine Learning: TensorFlow, YOLOv8, OpenCV, Scikit-learn, MobileNetV2, Computer Vision, Data Analysis

Web Development: HTML, CSS, JavaScript, Flask, REST APIs, SQLite

Tools: Git, GitHub, VS Code, MATLAB

PROJECTS

RISC-V Processor Design — RTL Design in Verilog

- Architected and verified a single-cycle RISC-V processor in Verilog HDL, covering arithmetic, logic, memory, and branch instruction classes.
- Built a Python-based assembler to convert assembly source into machine code, cutting manual test-vector prep time.
- Developed a full Icarus Verilog testbench suite validating datapath and control-unit correctness across all supported instructions, with complete architecture documentation.

Parameterized ALU — Verilog RTL Design

- Designed a parameterized ALU in Verilog with configurable bit-width, enabling reuse across datapaths of varying widths without rewriting RTL.
- Implemented 8+ arithmetic, logical, and shift operations behind a single opcode-driven interface; verified functionality through testbench simulation.

Relay — Multi-Model AI Chat Application

- Built a full-stack AI chatbot platform integrating multiple LLMs (DeepSeek, Groq, GPT) with dynamic model routing via REST APIs, allowing users to switch models mid-conversation.
- Migrated the platform to a SQLite backend with user authentication and persistent conversation history, turning a prototype into a stateful multi-user application.
- Delivered a full UI redesign and modular architecture that simplifies adding new model integrations.

AI Resume Analyzer

- Built a Python tool that parses PDF/text resumes and scores them against job descriptions for ATS compatibility using keyword-matching techniques.
- Designed a real-time upload interface that converts parsing output into actionable, scored improvement suggestions for users.

CERTIFICATIONS

- VLSI Design Fundamentals — HP
- Artificial Intelligence / Machine Learning — Simplilearn
- Python Programming — GeeksforGeeks

ACHIEVEMENTS & ACTIVITIES

- Completed Data Analytics (Tata, via Forage) and Project Management (Siemens, via Forage) job simulations.
- Competed in national-level hackathons Hackvyuha and InnoRise; advanced to elimination rounds in competitive Chess tournaments.

LANGUAGES KNOWN

English, Hindi, Kannada

HOBBIES

Chess, Art, Blogging, Writing